

Definition 1. A graph G is *bipartite* if its vertex set splits into two classes A and B such that every edge joins a vertex of A to a vertex of B .

Theorem 1. A graph is bipartite if and only if it contains no cycle of odd length.

Proof. One direction is immediate: walking around a cycle alternates between the two classes, so every cycle has even length. For the converse we may assume G is connected and fix a vertex r .

Claim 1.1. Colouring each vertex by the parity of its distance from r is proper.

Proof. Suppose an edge uv joins two vertices of the same parity. The shortest paths from r to u and to v together with the edge uv contain a closed walk of odd length, and every closed walk of odd length contains an odd cycle. This contradicts the hypothesis. $\square$

By Claim 1.1, the two parity classes witness that G is bipartite. $\square$

Theorem 2. Every tree with $n \geq 2$ vertices has at least two leaves.

Proof. Consider a longest path $P = v_0v_1 \cdots v_k$ in the tree.

1. The endpoint v_0 is a leaf.

Proof. A neighbour of v_0 outside P would extend P , and a neighbour on P other than v_1 would close a cycle. $\square$

2. By symmetry v_k is a leaf as well, and $v_0 \neq v_k$ because $n \geq 2$ forces $k \geq 1$.

$\square$

Theorem 3. For all $m, n \in \mathbb{N}$ the following hold.

- (a) $m + n = n + m$;
- (b) $m \cdot n = n \cdot m$.

Proof. We prove both parts by induction.

- (a) *Proof of (a).* Induct on n , using $m + 0 = m = 0 + m$ for the base case. $\square$
- (b) *Proof of (b).* Induct on n again, now using part (a) and distributivity in the step. $\square$

$\square$